

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

QUESTIONS: 5

Total Marks:20

Annexure A

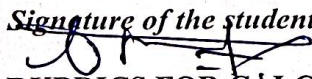
CLASS TEST

Criteria	Understanding of Task (2)	Procedure & Execution (2.5)	Observation & Result (2)	Viva Voce / Conceptual Response (2)	Time Management & Presentation (1.5)	Total(10)
Weightage	20%	25%	20%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Bhargesh A (192571006) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:


RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS:

Total Marks: 10x2 =20

Q1.

- a) Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]
- b) What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q2.

- a) Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]
- b). What is covariance analysis? List the two covariance formalms[L2, L3]

Q3.

- a) Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?
- b) Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q4..

- a) Find covariance for following data set $x = \{4,5,6,7,9\}$, $y = \{8,3,7,5,6\}$
- b) Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q5

- a). A hospital is analyzing patient age data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]
- b). Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, L5]

Data warehousing and Data mining

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CO2-AT-2 class-Test

81
100

① Data mining is the process of discovering useful patterns, relationship & knowledge from large datasets using statistical, machine learning & database techniques

Two Main tasks

KDD: knowledge discovery in (database) process

① Classification

② Clustering

① Data selection

② Data cleaning

③ Data integration

④ Data Transformation

Role of Data pre-processing

* Removes errors & inconsistencies

* Handles missing values

* Reduced Noise

* Improves data quality

(b) data cleaning: Is the process of detecting & correcting inaccurate, incomplete or inconsistent data

Two data Quality Issues & Handling Missing values

+ Missing values

+ Noisy data

+ Ignore records

+ Fill with mean, median or mode

Example

②

(a) Data Integration

+ Data integration combines data from different sources into one consistent dataset

Challenges

+ Data redundancy

+ Data inconsistency

Handling redundancy

+ Remove duplicate records

+ use correlation analysis

Handling inconsistency

+ Standardize formats

(b) Covariance Analysis

+ Covariance measures how two variables change together

+ Positive covariance $\rightarrow$ Increase together

+ Negative covariance $\rightarrow$ Move in opposite directions

Q 2)

$$A = (2, 3, 5, 4, 6)$$

$$B = (5, 8, 10, 11, 14)$$

$$\text{Sample covariance} = 5$$

* Since covariance is positive both stock prices generally rise or fall together.

(b) Architecture of data mining system

Data source $\rightarrow$ Database / Data warehouse $\rightarrow$

Data preprocessing $\rightarrow$ Data mining engine $\rightarrow$

Pattern evaluation $\rightarrow$ Knowledge presentation

$\rightarrow$ user

Components

- * Data source collect data
- * Data warehouse stores data
- * Preprocessing cleans data
- * Mining engine discovers pattern
- * Pattern evaluation selects useful results

Q (a)

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 7, 7, 5, 6\}$$

$$\text{Sample Covariance} = -0.45$$

* The negative covariance indicates a weak negative relationship between x and y

(b) Steps

- ① Data cleaning
- ② Data Integration
- ③ Data Transformation
- ④ Data reduction
- ⑤ Data Discretization

Data Preprocessing

Pipeline

- * Remove Duplicates
- * Fill missing value
- * Remove outliers
- * Standardize formats.

(a) using equal-frequency (equal-depth) binning with 3 bins:

* Bin 1 Mean = 18

* Bin 2 Mean ≈ 28.11

* Bin 3 Mean ≈ 43.78

These mean values replace the original values in each bin to smooth the data.

(b) Data Discretization

* converts continuous data into interval classes

Importance

- * Reduce noise
- * Simplifies data
- * Improves mining performance

} Data Discretization }

Equal-width	Equal-frequency
Equal interval size	Equal number of records
Simple	Better for skewed data
Fixed width	Variable width